

 OR-AS Operations Research Applications and Solutions	Case Name: Apartment Building Finishing Works (3)	Sector	Construction (residential building)	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
			One of nine std. scenarios	
Submitted by	William & Jef		User defined distributions	
Date	2016	Project Control	Automatic tracking	
File Name	C2016-33Apartment Building Finishing Works (3)		Tracking based on user input	

1. Project description

Project authenticity

The project concerns the finishing works of an apartment building, planned from March to August 2016 with a total budget of €377,282 over 116 working days. It comprises two main sections, ground floor and level +1, covering plastering, flooring, carpentry, painting, and furniture installation. Four tracking reports show that the project was completed later and at a higher actual cost (€394,829) than originally budgeted..

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	24	Serial/Parallel (SP)	50%
Planned Duration (PD)	116	Activity Distribution (AD)	72%
Budget At Completion (BAC)	377,282.00€	Length of Arcs (LA)	0%
Renewable Resources	-	Topological Float (TF)	30%
Consumable Resources	-		

* standard eight-hour working days=

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0.00	0.00	N/A
CRI-rho	100.00	0.00	N/A
CRI-tau	100.00	0.00	N/A

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	-	-	-
CRI-rho	-	-	-
CRI-tau	-	-	-

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	43	47	0.30
SI	46	38	0.38
SSI	14	18	1.00
CRI-r	16	16	1.08
CRI-rho	17	15	1.06
CRI-tau	11	10	1.13

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	2.17	1.01	1	0.00	0.00
PV - SPI	2.69	1.13	CPI	0.00	0.00
PV - SCI	2.69	1.13	SPI	0.57	0.12
ED - 1	2.10	0.97	SPI(t)	10.54	-9.65
ED - SPI	2.62	1.06	SCI	0.57	0.12
ED - SCI	2.62	1.06	SCI(t)	10.54	-9.65
ES - 1	4.21	-2.61	0.8 CPI + 0.2 SPI	0.11	0.00
ES - SPI(t)	15.14	-12.67	0.8 CPI + 0.2 SPI(t)	1.37	-1.15
ES - SCI(t)	15.14	-12.67			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 4 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-9965.00	-58960.00	-7.75	0.95	0.73	0.91	1
std dev	5319.49	48014.26	3.86	0.03	0.20	0.01	0
final	-17547.00	0.00	-13	0.96	1.00	0.90	1

2.3.3.2. Time forecasting

PD	116	Real Duration	129	Late	11.21%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	186.23	20.27	44.36	-44.36
PV - SPI	231.97	60.40	79.82	-79.82
PV - SCI	246.91	70.86	91.40	-91.40
ED - 1	188.58	9.13	46.19	-46.19
ED - SPI	236.22	54.04	83.12	-83.12
ED - SCI	245.04	65.57	89.95	-89.95
ES - 1	171.13	5.44	32.66	-32.66
ES - SPI(t)	177.29	1.91	37.44	-37.44
ES - SCI(t)	182.22	9.30	41.25	-41.25

2.3.3.3. Cost forecasting

BAC	377,282.00€	Real Cost	394,829.00€	Over budget	4.65%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	387247.00	5319.49	1.92	1.92
CPI	398910.11	13352.07	1.99	-1.03
SPI	493718.11	122146.36	25.05	-25.05
SPI(t)	403425.09	13281.69	2.31	-2.18
SCI	514446.84	152231.80	30.30	-30.30
SCI(t)	416414.83	31690.34	5.47	-5.47
0.8 CPI + 0.2 SPI	411690.22	26608.16	4.50	-4.27
0.8 CPI + 0.2 SPI(t)	399784.77	13247.17	1.86	-1.26